

Essentialism and the Essence of Being Necessary

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In his 1994 paper *Essence and Modality* Kit Fine argued against the view that essence should be understood in terms of *de re* modality. He suggested the conversed account: the metaphysical necessity should be understood in terms of essence, of what is true in virtue of the nature of what. The broad tendency to explain modality in terms of essence was followed by philosophers such as Hale and Lowe, albeit some differences in details. I will classify all such accounts as *essentialism* on modality.

An obvious challenge for essentialists would be to provide an account on essence that does not rely on modal notions. And at the same time, they also should derive the fact that p being necessarily true from the fact that p being essentially true. Skeptics argue that this was never done. The arguments provided for the latter implicitly assumes some modal quality on the essentiality of p , and that is to fail to satisfy the former requirement. And if the essentiality of p has no modal quality at all, there seems to be a gap between something being essential and being necessary, so that the derivation from essentiality to necessity is impossible to achieve.

In this paper, I will argue that essentialists can reply to the skeptic challenge without compromising any of their core thesis. The skeptics overlook one remaining possibility: that the gap is filled by the essence of *being necessary*. More specifically, I suggest that both requirements can be satisfied by assuming that it is true in virtue of the nature of being necessary that if something is essentially true, then it is necessarily so.

In section 1, I will summarize the problem of the source of necessity, and how essential-

ist answer was suggested as a *non-transmissive* explanation on necessity. I also argue that *all* essentialist explanation would be non-transmissive. The skeptic challenge mentioned above will be summarized in detail in section 2. I will argue in section 3 that any attempts on non-transmissive explanation would ultimately appeal to the account on the essence of being necessary, including of course essentialism. And then I will also show how the criticisms can be replied by doing so in section 4. In section 5, I will clarify why it is legitimate for essentialists to take it is true by the nature of being necessary that if something is essentially true, then it is necessarily so. Especially, I argue that it is not circular, at least viciously. Note that I am not trying to argue for the essentialist answer for the problem of the source of necessity, against other alternatives.

1 The Source of Necessity and Non-Transmissive Explanations

For any necessary fact p , we can ask *why* is p necessary. The question is different from the question of why does p obtain. For example, we can answer the question of why is it the case that $3^2 = 9$ or snow is white by saying that it is the case that $3^2 = 9$. But this cannot be the answer for the question of why is it necessarily the case that $3^2 = 9$ or snow is white.

Similarly, the question of the source of *particular* necessity should be distinguished from the question of the *general* source of necessity. It might be possible to address the source of the particular necessity of the previous fact by the necessity of $3^2 = 9$. This however does not guide us at all about the general source of necessity, since the necessity of $3^2 = 9$ is left unexplained. The general source of necessity should be able to provide, mediately or immediately, the source of all necessity.

One difficulty answering the question is noted by Blackburn (1993). He argued that the general source of necessity cannot be explained by some true propositions. Suppose we want to answer the question by pointing to some true propositions, that is, to claim that $\Box p$ because q . It would be obvious that q is either necessarily true or contingently true. If it is necessarily

true, then it cannot be the final source of necessity, for it leaves the necessity of q unexplained. But it also cannot be the case that q is contingently true, for “... then there is strong pressure to feel that the original necessity has not been explained or identified, so much as undermined” (Blackburn 1993, p.53)

Blackburn wishes to further conclude that therefore any realist conception on necessity must fail. This advance might be too hasty. One can insist that necessity is metaphysically primitive, therefore not explainable by other notions. It also might be suggested that the source of necessity need not be truth-bearers such as propositions. Not every possibility is closed by the proposed dilemma.

However, it is also argued that to explain necessity via contingently or necessarily true proposition is indeed possible. Hale (2002) pointed out that the denial of contingently true propositions explaining necessity presupposes that necessity is never contingent, which some philosophers, notably conventionalists would disagree. Still, Hale himself does not try to locate the source of necessity among contingent truths. He advocates what he calls “non-transmissive” explanation on modality. One example of a transmissive explanation of modality is mentioned earlier: $\Box(3^2 = 9 \text{ or snow is white})$ *because* $\Box 3^2 = 9$ and $\Box p \rightarrow \Box(p \vee q)$. Here not only the truth of $3^2 = 9$ but also the necessity of it is used for the explanation, i.e., the explanation points out that the necessity is transmitted from the explanans to the explanandum. In general, a transmissive explanation will take the form of $\Box p$ *because* $\Box q$, where q itself cannot explain $\Box p$. Perhaps Blackburn is correct if this is the only available form of explanation. It may help us to sort out some “fundamental” necessities from non-fundamental ones, but does not give us the general source of necessity, for it is precisely the fundamental necessities that needs to be accounted their sources.

Non-transmissive explanations, on the other hand, take the form of $\Box p$ *because* q , where q is a proposition that is necessarily true “but in which what explains the necessity of the explanandum is not q ’s *necessity*, but its *truth simpliciter*.” (Hale 2002, p.311) Contra Blackburn, Hale explicitly takes some necessarily truths to be the general source of necessity. Since q is also a necessary truth, the necessity of $\Box q$ would also be explained via the truth of a proposition

which is necessarily true, say r . But that does not imply that the necessity of p is better explained by r . For, it is the truth simpliciter of q that explains the necessity of p , not its necessity. Rather, if the truth of q is explained ultimately by the truth of s , it would make sense to say the necessity of p is explained ultimately by s .

Therefore, if we succeed in giving a non-transmissive explanation to all, or at least the fundamental necessities, it will suffice for being an account for the general source of necessity, which does not appeal to further necessities. This is usually done by specifying what kinds of necessary truths explains the related necessities. Hale's preferred view is that "Necessarily p because $\Box_{X_1, \dots, X_n} p$ " (Hale 2013, p.152): the necessity of any proposition p is explained by it being true in virtue of, or being essentially true of some objects $X_1, \dots, X_n$. But this is not the sole possible explanation. For instance, modal realist would assert that the necessity of any proposition p is explained by p being true at all possible worlds. Now, if the notion of possible worlds can be specified without modal notions, this is a genuine case of a non-transmissive explanation: it is necessary that p is true at all possible worlds (given that p is a necessary truth), but the truth of p being true at all possible worlds, not its necessity, does the explanatory work. I will classify all those theories of the source of necessity which utilizes non-transmissive explanation as "non-transmissivism".

Before proceeding, let me clarify between various senses and kinds of essence. In most cases, by "essence" I mean *mediate and consequential* essence, following Fine (1995a). Suppose we have a clear idea of a strict and direct essence, which is immediate and constitutive. Mediate essence of F then additionally includes the nature of things that are involved in the nature of F ; consequential essence of F includes what logically follows from the nature of F . I always include mediate essence throughout the paper, but I will occasionally restrict myself to mediate and constitutive essence. I will make myself clear when the shift occurs.

Also crucial is the distinction between objectual essence and predicational essence. There is the essence of some particular object, the nature of the thing in question; and the essence of being such and such. Fine (2015) argued the essence of being F is different from the essence of the concept F . Suppose the objectual essence of the concept *concept* that it is itself a concept:

$\Box_{concept}(Concept \text{ is a concept})$. Then it is a consequential essence of the concept *concept* that $\exists x(x \text{ is a concept})$, that there is something that is a concept. However, it seems quite implausible to think that it is by nature of *being a concept* that there is something satisfies it. That is, while $\Box_{concept}\exists x(x \text{ is a concept})$ seems plausible, $\Box_{being-a-concept}\exists x(x \text{ is a concept})$ does not (p.298). This, we may say, depicts the difference between the objectual essence of concept and predication essence of being a concept.

It would be clear that the essence of objects or things would be the objectual essence of that object. And following Fine (1995a, 2015), I will reduce objectual essence to the predication essence of *rigid predicate*, which denotes the property of being identical to some object. Therefore the objectual essence of *a* would be the same as the predication essence of $\lambda x.x = a$. I will occasionally write $\Box_a$ to denote the objectual essence of *a*, but that would merely be the abbreviation of $\Box_{\lambda x.x=a}$. Finally, “the (objectual) essence of objects” would mean the same thing as “the (predication) essence of rigid predicates”.

It is reasonable to expect any sensible essentialist account on modality would be non-transmissivist. Essentialists typically accept the biconditional $\Box A \leftrightarrow \exists F \Box_F A$ (Fine 2020, p.461): a sentence or a proposition is necessary if and only if it is true in virtue of the nature of some objects or properties. And since essentialists typically holds that the necessity of *A* is explainable by it being true in virtue of the nature of some *objects*,¹ it is a theorem under the standard axioms of the logic of essence that $\Box_F A \rightarrow \Box_F \Box_F A$, where the predicate *F* is rigid, i.e., denotes a property of being identical to some object (Fine 1995a, p.255, see THEOREM 6 (ii)). That is, if *A* is true in virtue of the nature of *F*, the fact that *A* is true in virtue of the nature of *F* is true in virtue of the nature of *F*. It implies that, since essentialists accept the biconditional above, $\Box_F A$, the explanan of the necessity of *A*, must also be necessary. But since essentialists would want $\Box_F A$, not the necessity of $\Box_F A$ to explain the necessity of *A*, it constructs the exact case of the expalnation of necessity where the explanans are necessarily true yet its *truth* simpliciter, not necessity explaining the necessity of the explanandum.

¹Fine (1994) writes: “The metaphysically necessary truths can then be identified with the propositions which are true in virtue of the nature of all *objects* whatever” (p.9, italics added); Hale (2013) also writes: “Metaphysical necessities might plausibly be identified as those which hold in virtue of the nature of *things* in general.” (p.150, italics added). Additionally, the variable *X*, although not formalized, are almost always introduced explicitly as variable ranging over various kinds of entities.

Note that there can be slightly different view on how essence can account for the source of necessity among essentialists. For Kit Fine, being metaphysical necessary is *defined* to being true by the the essence of all objects (Fine 1994, 1995a). The notion of metaphysical necessity is therefore reduced in terms of essence. This implies that Fine would agree that the necessity of A would be grounded on A being true in virtue of the nature of all objects. Following the notation presented in Fine (2015), such a definition which gives necessary and sufficient condition of being necessary yields $\Box_{\forall} A \leftrightarrow_A \Box A$, where $\forall$ is defined as $\lambda x(\Pi x(x = x))$, and Π is a quantifier ranging over all possible objects. That means that something being true in virtue of the nature of all objects is both (partial) ground and essence of something being necessary. Hale, on the contrary, explicitly claims his account to be non-reductive. Although he agrees with Fine that metaphysical necessity is grounded in the nature of things, he also takes modality as “fundamental and irreducible” (Hale 2013, Ch.3). Lowe (2008) also suggests that all metaphysical necessity is grounded on essence, where essence just is the nature of an object, not a further entity. It is not very clear whether Lowe would agree that necessity can be defined in terms of essence, but it seems likely that Lowe would not equate metaphysical necessity and the maximal essence as Fine does. For, it is doubtful whether Lowe would include mediate and consequential essence (cf. Fine 1995c) when he speaks of essence.

2 The Gap between Essence and Being Necessary

One restriction for all essentialists is that the notion of essence should not be understood in terms of modal notions. For example, if the definition of Fa being true in virtue of the nature of a is just that a necessarily having F , then the source of necessity of Fa cannot be that Fa being true in virtue of the nature of a , for that would still leave *de re* necessity unexplained, or ending up being circular. It is parallel to the reason why possible world theorists should not define possible worlds as worlds (mereological aggregates) that are metaphysically possible. But if essence should be understood independently from necessity, how can one ensure that all essential facts are necessary?

It seems that essentialists must accept that all essential truths, what is true in virtue the nature of something are necessary. To acknowledge contingent essence, one must give a criteria demarcating them with necessary essence without appealing to the notion of essence. That is, one must identify the non-essentialist source of the necessity of necessary essences, which contingent essences lack. And this apparently undermines essentialism.

Some skeptics question exactly that whether all essential truths are necessary. To put it formally, essentialists are committed into what Mackie (2020) calls “Necessity Principle (NP)”:

$$(NP) \quad \forall F \forall A (\Box_F A \rightarrow \Box A)^2$$

where F is any rigid predicate and A is any sentence. I will occasionally omit one or two universal quantifier for convinience and readability. But what ensures (NP)? One common understanding of essence among essentialists is by *real definition*: roughly, the essence, or the identity of an object is captured or given by its real definition, and therefore what is true in virtue of the nature of something is what is true by the definition of something (Fine 1994, p.2; Hale 2013, p.132; Lowe 2012, p.919). Alternatively, Correia and Skiles (2017) suggest to understand essence in terms of generalized identity, expressed by the identity statements such as “for a thing to be a bachelor is for it to be an unmarried adult male” or “for the Atlantic Ocean to be filled with water is for it to be filled with H_2O molecules” (p.643). Here being F expresses the full essence of being G if and only if to be G is to be F ; it expresses partial essence of being G if and only if there is some H such that to be G is to be both F and H (pp.649-50). The skeptic charge is that there seems no direct evidence that the nature of real definition or generalized identity can ground (NP) alone. How can essentialists “deliver a modal rabbit out of a non-modal hat” (Mackie 2020, p.252)?

In the case of definitional essentialists, Hale tried to give an argument. Suppose $\Phi\alpha$ is true in virtue of the identity of α . Now suppose it is possible that $\neg\Phi\alpha$, which follows that there is a β such that $\alpha = \beta \wedge \Diamond\neg\Phi\beta$. But β cannot be identical to α for it does not satisfy the condition to be identical to α , which is Φ . Therefore $\neg\alpha = \beta$, contradiction. Since contradictions are

²This formalization follows Fine (2020).

impossible, we can conclude that if $\Phi\alpha$ is true in virtue of the identity of α , $\Phi\alpha$ is necessary (Hale 2013, p.133).

But it has been pointed out that the argument does not successfully answer to the skepticism toward (NP) (Romero 2019, pp.125-7; Leech 2021, pp.893-5). For the proposed argument to succeed, it must be assumed that Φ is *necessarily* the condition for some arbitrary objects to be identical to α . Either $\Box(\Box_\alpha\Phi)$ or $\Box\Phi\alpha$ is required. But what is assumed is only that Φ is *actually* the condition of the identity. To hold that the assumption implies $\Box\Phi\alpha$ simply begs the question. And while $\Box_\forall(\Box_{\lambda x.x=\alpha}\Phi)$ is provable from $\Box_{\lambda x.x=\alpha}\Phi$, the inference by Finean definition from $\Box_\forall A$ to $\Box A$ again begs the question, for that is just another form of (NP). Moreover, Hale himself would not agree to the identification between necessary truths and the proposition true in virtue of the nature of all objects.

In fact, one can construct a proof for (NP) once she accepts the Finean definition of necessity. It is an axiom in the logic of essence that $F \subseteq G \rightarrow (\Box_F A \rightarrow \Box_G A)$, where $R \subseteq G$ is for $\Pi x(Rx \rightarrow Gx)$ (Fine 1995a, pp.246-7). Since for any rigid predicate F $F \subseteq \forall$, it follows that $\Box_F A \rightarrow \Box_\forall A$. And since the necessity operator $\Box$ is just defined as $\Box_\forall$, we can derive $\Box_F A \rightarrow \Box A$, which is (NP) itself. But there is another reason why skeptic regard the definition to be unsatisfactory (Romero 2019, p.129-36). As it was mentioned, Fine defines necessary truths as propositions that are true by the essence of all objects. He writes $\Box_\forall S$ for S being essentially true in virtue of all objects, where $\forall$ is defined as $\lambda x(\Pi x(x = x))$, and Π is a quantifier ranging over “all metaphysically possible objects.” (Fine 1995a, p.250). Persuasively, since the problem of which objects are possible depends on the notion of modality, to conceive necessary truths as maximal essence, the essence of all *possible* objects, is circular. In the same paper Fine also writes “[...] one might operate with a broader conception of possible object than is customary in modal logic” (ibid, p.244), but it is still unclear what the “broder conception of possible object” is.

To understand essence in terms of generalized identity may seem to have better chance for one to advocate (NP). “As with objectual identity, generalized identity is tightly linked to metaphysical necessity” (Correia and Skiles 2017, p.656). Following Correia and Skiles, let “ $p \equiv q$ ” denote the generalized identity between two predicates or propositions p and q . As it is the

case of the objectual identity, $p \equiv p$ is a theorem³ and therefore in modal logic it follows that $\Box(p \equiv p)$. And given $p \equiv (q \wedge r)$ [$p \equiv q$], which means that q is a partial [full] essence of p , by substitutability we get $\Box(p \equiv (q \wedge r))$ [$\Box(p \equiv q)$]. That is, assuming q is a partial [full] essence of p , it follows that *necessarily* q is a partial [full] essence of p .

This actually proves something stronger than (NP): namely, $\Box_F A \rightarrow \Box \Box_F A$. However, since if something is true in virtue of the nature of something it would also be true *simpliciter*, it would be reasonable to think that implies (NP). The problem is that the proof presupposes $\Box(p \equiv p)$; that the generalized identities are necessary. How can essentialists explain that particular necessity? It was argued that the necessity of $\Box(p \equiv p)$ cannot be explained by the essence of generalized identity (Leech 2021, pp.901-2; cf. Romero 2019, p.138). For example, suppose that the necessity is explained by that for two things to be generically identical is partially to bear a reflexive relation. Grant, but what explains the necessity of the partial generalized identity between generalized identity and reflexive relation? Here it seems we cannot appeal to the necessity of $p \equiv p$, for that exactly is what it is explaining, not something that is explained by. Therefore, to appeal to the essence of generalized identity itself cannot serve as the final source of the necessity of all generalized identity.

Leech seems to conclude that, in explaining the necessity of the generalized identity that expresses the (partial) essence of generalized identity itself “*one should not appeal to the essence of anything to account for this necessity*”. She further considers this constraint as “the crucial and general objection” on the necessity of essence (p. 902).

I disagree. I think Leech overlooks one remaining possibility of appealing to the essence: that generalized identities, including $p \equiv p$ are necessary because of the essence of *being necessary*. Furthermore, in my view, this also applies directly on (NP). Every propositions true in virtue of the nature of something or all things are also necessarily true, not because of the nature of essence but because of the nature of what is it to be necessary. In section 2, I will argue that essentialists must accept that (NP) is true in virtue of the nature of what is it to be necessary. It will also be shown that challenges presented in this section can be suitably replied

³For they take generalized identity to be reflexive (p.645)

once we notice that (NP) is true in virtue of the nature of being necessary. In section 3 I will argue that this is legitimate.

3 Non-transmissive Explanations and the Essence of Being Necessary

I argued that any sensible essentialist account on necessity would hold that essence provides a non-transmissive explanation on necessity. Non-transmissive explanations, as mentioned, take the form of $\Box p$ because q , where q is also necessarily true. Usually, the role of q will be taken by a proposition *about* p . For example, that *p is true in all possible worlds*, or that *p is true in virtue of the nature of something*, and so on. Let C denote some such status, or criteria that a proposition can satisfy. A theory on the general source of necessity will then give an account of what the status C is. Then we can modify the form of non-transmissive explanation into $\Box p$ because $C(p)$, where p necessarily satisfies C . Essentialists are those who hold that the status C is being true in virtue of the nature of something.

It is obvious that for any non-transmissivist account, which takes such C to be the general source of necessity, must also hold that C can be understood without appealing to modal notions. But one must also affirm that $C(p) \rightarrow \Box p$ for any proposition p , for the same reason why essentialists must accept (NP). If there are some true propositions that satisfies C but not necessarily true, it means that there are something additional to C that makes a propositions satisfying C necessary. C is then at most only a part of the criteria a proposition must satisfy to be necessary, contrary to the initial assumption.

Therefore, the problem proposed by skeptics are not peculiar to essentialists; it turns out to be a general difficulty for any non-transmissivist account. For example, a modal realist would say that $\Box p$ because p is true in all possible worlds. But if the notion of possible worlds is defined without modal notions, one can equally ask how modal realists can “deliver a modal rabbit out of a non-modal hat”. The general difficulty is to provide an elucidation of how a non-modal understanding of C can get along with the requirement of $\forall p(C(p) \rightarrow \Box p)$. I will

call this conditional “General Necessity Principle (GNP)”, for it is analogous with (NP).

How can a non-transmissivist argue for (GNP)? One thing to notice is that one *cannot* merely appeal to some fact about C or necessity that is not essential to being C or being necessary, but take $C(p)$ to be the ultimate source of $\Box p$. Suppose a proposition satisfying C is necessary as long as the criterion C satisfies other non-essential criteria, say X . That is, (GNP) depends on $X(C)$, but it is not a part of the nature of being C that $X(C)$. Then, it seems what *explains* $\Box p$ is not merely $C(p)$, but that $C(p) \wedge X(C)$, since $C(p)$ cannot guarantee $\Box p$ alone. Additional component of $X(C)$ is required because the nature of something being C does not by itself ensure that. So the account that $\Box p$ because $C(p)$ fails, for being C turns out not to be everything that is needed to explain necessity. Similar argument also undermines the option of appealing to non-essential fact about being necessary. Suppose (GNP) obtains as long as $Y(N)$ obtains, where N is the property of being necessary⁴, and Y is a non-essential property of being N . Again, then the correct account should be that $\Box p$ because $C(p) \wedge Y(N)$, and therefore the original account turns out to be untenable. To sum up, if one wants to insist that $C(p)$ provides full explanation for $\Box p$, one cannot think (GNP) is true in virtue of some facts about being C or being necessary that is not essential to them.

The previous argument does not show it cannot be the case that $X(C)$ or $Y(N)$ grounds (GNP); it shows that even if one believes that $X(C)$ or $Y(N)$ grounds (GNP), to be a non-transmissivist one should also take (GNP) to be true in virtue of the essence of C or N .⁵ However, it seems that non-transmissivists also cannot take (GNP) to be true in virtue of the nature of C . Suppose for the contrary that it is. That is, suppose $\Box_C(C(p) \rightarrow \Box p)$. Now, if one takes the notion of essence as closely tied to the notion of real definition (Fine 1994), it follows that the criteria C is, partially, *defined* in a way such that it is the case that $C(p) \rightarrow \Box p$. Or following a revised version of definitional essentialism (Fine 2015), it follows that $C(p) \rightarrow \Box p$ is *constitutive* necessary condition for some p to be, to satisfy C . This seems to me as straightforwardly suggesting that C should be understood partially in terms of necessity.

⁴Here I ignore the problem between the predicational and operational view of necessity. The predicate N can be regarded as being defined as $\forall p(\Box p \leftrightarrow N(p))$.

⁵By ‘essence’ here I *include* mediate essence, but not consequential essence. See Fine (1995c)

Perhaps the argument can be made clearer by introducing Fine's notion of ontological dependence. According to his account on ontological dependence (Fine 1995b), x depends on y "if y is a constituent of a proposition that is true in virtue of the identity of x " (p. 275). Therefore from $\Box_C(C(p) \rightarrow \Box p)$, it follows that C ontologically depends on the modal operator $\Box$. Fine's notion of ontological dependence is not antisymmetric, therefore the possibility of the necessity depending upon C is still open. But what non-transmissivists need is not a mere dependency, but that C is strictly prior to necessity which prevents C being defined, or constituted in terms of necessity.

The only remaining possibility, then, is to say that (GNP) is true in virtue of the nature of being N . Any non-transmissivist account on necessity must take their relevant instances of (GNP) to be true because of the essence of being necessary. And in fact, although not explicitly stated, Kit Fine's account implies that. As I mentioned earlier, Kit Fine *defines* being necessary in terms of being true in virtue of all things: $\Box_{\forall} A \leftrightarrow_A \Box A$. And its implication $\Box_{\forall} A \leftarrow_A \Box A$ is just another, intuitive way to put $\Box_N \forall A (\Box_{\forall} A \rightarrow \Box A)$ (Fine 2015, p.299). For any F $\Box_F A \rightarrow \Box_{\forall} A$, and that is probably true in virtue of the nature of some relevant logical notions. Since from $\Box_{\forall} A \rightarrow \Box A$ and $\Box_F A \rightarrow \Box_{\forall} A$ it follows that $\Box_F A \rightarrow \Box A$, it therefore is a mediate essence of being necessary that $\forall A (\Box_F A \rightarrow \Box A)$ - which is (NP).

But I also mentioned that such a definition is accused to be circular, for the complex property $\forall$ presupposes the domain of *all metaphysically possible* object. An additional concern is that we might want to talk about the essence of *impossible* objects, which are not in any broad sense of "possible object" one of them. Typical examples are the Russellian set or the biggest prime number. Why is the Russellian set impossible? An essentialist line of argument might be something like the following. The Russellian set is essentially a set of all sets that does not contain itself; therefore it is necessary that for every x , if x is identical to the Russellian set, it is a set of all sets that does not contain itself. And it is also necessary that, for any x such that it is the set of all sets that does not contain itself does not exist. Therefore, Russellian set necessarily does not exist, i.e., is an impossible object. But for such an argument to be sound, it should be held that the essence of an impossible object can also be the source of some necessities.

If essentialists include the essence of impossible objects as the source of necessity, how can the Finean definition that to be metaphysically necessary is to be the essence of *all* objects be maintained? One way out would be to construct a domain which includes both possible and impossible objects. But which objects, possible or impossible, are out there? Perhaps Meinongians could give us an answer.⁶ However, it seems to me as a unnecessary detour to think that essentialists would need a theory of objects.⁷ It is not very desirable for essentialists to just wait until object theorists come up with a comprehension principle of object every essentialists can happily endorse.

Instead, I suggest to take $\Box_N(\Box_F A \rightarrow \Box A)$ not as a consequence of the definition, but directly into the essentialist account on necessity. Here again N denotes the property of being necessary; F is any rigid predicate, including predicates on identity with impossible and possible objects, and A is any sentence. Following definitional conception on essence, that amounts say that (NP) is part of the definition of *being necessary*, not the definition of *being essentially true*. The Finean intuition that to be necessary is to be true in virtue of the nature of *all things* would then be preserved under any suitable theory of objects, as long as the axiom $F \subseteq G \rightarrow (\Box_F A \rightarrow \Box_G A)$ is employed.

Since essentialists are one specific kind of non-transmissivists, not only Fine but all essentialists must accept $\Box_N(\Box_F A \rightarrow \Box A)$. Otherwise, essence turns out not to provide the full explanation for necessity; or the disturbing consequence that being essential ontologically depends on being would follow. Therefore, any compelling argument for essentialism would also be the argument for that (NP) is true in virtue of the nature of being necessary. Competing theories then would be some other non-transmissivist accounts. For example, modal realists, since there are also non-transmissivists, will and must argue it is rather that *if something is true in all possible worlds then it is necessary* is the partial essence of being necessary. For I do not aim in this paper to argue for the essentialist answer for the problem of the source of necessity

⁶See, for example Zalta (1983) or Priest (2005). Fine himself also had worked on the problem of non-existent and the comprehension principle on objects. The negative part, which he criticizes certain kinds of object theories was published in Fine (1982). There he also stated that the positive project where he develops his own theory will be published in part II of the paper, but unfortunately it remains as a non-existent object.

⁷It is even unclear if an independently developed object theory would get along with essentialism well. For example, Zalta, although I see no logical inconsistency between his theory of objects and essentialism, advocates a modal conception of essence specified in terms of his theory of objects in Zalta (2006).

against other alternatives, I will not explore the possible arguments for the thesis of (NP) as the essence of being necessary.

4 How to Fill in the Gap

What I *should* show in this paper, is how essentialists can answer the challenge raised by Romero, Leech and Mackie by locating (NP) to the nature of being necessary. To briefly summarize, their main criticism was that once we stick to the notion of essence that is *not* defined in terms of necessity, there is a gap between essence and necessity - there seems to be no good reason to think all essential truths are necessary. That is, we cannot ensure (NP) by non-modal conception of essence, which seems to be crucial if essentialists want to explain all necessity by essence. My suggestion is: the gap is to be filled in by the nature of necessity, not anything else such as the definitional notion of essence or generalized identity.

Let us return to Hale's argument for the necessity of definitional essence. I take Romero's criticism to be partially correct, to the extent that we cannot infer $\Box\Phi\alpha$ from $\Box_\alpha$ without taking the essence of being necessary into the argument. We cannot appeal only to the immediate essence of α to establish $\Box\Phi\alpha$. So there indeed is a gap between the essence of an objects and *de re* necessity. The gap, however, can be filled in by the nature of being necessary. So the revised argument goes as follows. Suppose again, that $\Box_\alpha\Phi\alpha$. Now, Romero and Leech ask: why β such that $\Diamond\neg\Phi\beta$ cannot be identical to α , if the initial presupposition was only that it is actually $\Box_\alpha\Phi\alpha$? Now we have a further reply for that. Since by the definition of $\Diamond$, it is equivalent to $\neg\Box\Phi\beta$ ⁸. Suppose for contradiction, that $\alpha = \beta$ and therefore $\neg\Box\Phi\alpha$.

Now, from $\Box_N\forall F\forall A(\Box_FA \rightarrow \Box A)$ we can prove, utilizing the *T-axiom*: $\Box_FA \rightarrow A$ (Fine 1995a, pp.247-8) and modus ponens that $\forall F\forall A(\Box_FA \rightarrow \Box A)$. By universal instantiation we get $\Box_\alpha\Phi\alpha \rightarrow \Box\Phi\alpha$; and by the assumption $\Box_\alpha\Phi\alpha$ and modus ponens we get $\Box\Phi\alpha$. Contradiction.

⁸Perhaps one would prefer to have $\Diamond$ as primitive and $\Box p$ to be defined as $\neg\Diamond\neg p$. It is certainly possible for essentialists to follow that: the source of *possibility* of any possible truths q is explained by there being no rigid predicate F such that $\neg q$ is true in virtue of the nature of F . Alternatively, one can say it is explained by there being an object that does not have $\neg q$ as a mediate and consequential essence. Any other arguments surrounding essentialism can be reconstructed in such way.

Therefore, there can be no β such that $\alpha = \beta \wedge \Diamond\Phi\beta$, which ensures that if Φ is actually an essential property of α , then it is a metaphysically necessary condition for something to be identical to α .

It is important to notice that the argument depends on the essence of being necessary, and the logical rules of the essentialist operator, the universal quantifier and the material conditional. So we do not suppose essential truths are not *ab initio* necessary. The necessity is to be derived from the essence of being necessary and the relevant logical rules⁹. I will argue that it is legitimate to appeal to the essence of necessity to validate essentialism in section 5.

This is nevertheless not to ignore the essentialist intuition that it is in some sense “obvious” that essential facts are necessary. I suggested that essentialists cannot take essential fact to be *ab initio* necessary nor take it to be necessary in virtue of the nature of some facts being essential. However, it seems also true that there certainly is some kind of frustration when essentialists are confronted with questions like “why are essential facts necessary?” or “why an object necessarily has the property that defines what the object is?” This is, I suggest, same kind of frustration when one is confronted with questions like “why is water H₂O?” or “why does Socrates” singleton set have Socrates as member?”. According to Glazier (2017), these questions are subject to essentialist explanation. We can simply answer: “that is just what water is” or “that is just what the set is”. Essentialists therefore can take $\Box_N(\Box_F A \rightarrow \Box A)$ as an explanans of such an ultimate essentialist explanation of (NP). “Why are essential facts necessary?” I suggest one kind of answer essentialists can give is: “that is just what necessity is!”

So much for the definitional conception of essence. We can accept a proposition is necessarily true because it is true in virtue of the nature of *some* - instead of *all* - objects. Being essentially true guarantees being necessarily true partially because it is the essence of *being necessary* that if a proposition is true in virtue of the nature of some objects, then it is necessarily true. To vindicate the necessity of a particular essential fact, one must appeal to the essence

⁹Which is common to expect to be true in virtue of the nature of the relevant logical operators. For example, Fine (1994) writes “the logical necessities can be taken to be the propositions which are true in virtue of the nature of all logical concepts”. At least every instances of the logical rules would be true in virtue of the nature of logical concepts, and even the rule itself if it can be taken as a higher-order proposition.

of being necessary and other logical rules.

What if essence is to be understood in terms of generalized identity? One immediate idea from the previous observation might be that: $\exists q(Np \equiv_p (\forall x \exists r([\lambda y.y = x]z \equiv_z (p \wedge r))) \wedge q))$. This means that it is a partial essence of being necessary that: if p is an partial objectual essence of any x , then p is necessary. To be necessary *just is* to be an partial essence of some object. We can now proceed in a simillar way as we did in the case of definitional essence. How can we argue here that therefore all objectually essential facts are necessary? Suppose we have an statement of objectual identity: $[\lambda y.y = a]x \equiv_x (s \wedge w)$. In other words, s is a partial essence of the object a . Since it follows from the nature of being necessary that for s to be necessary is just to be a partial essence of a , from the previous statement we can infer that s is necessary.

Interesstingly, here we have another option. Remind in section 2, the discussion on the essentialism in terms of generalized identity were questioning not (NP) but the necessity of the generalized identity: for any p , $\Box(p \equiv p)$. Can we account for that instead? I think yes. Since we cannot say $\Box(p \equiv p)$ is true by the essence of the generalized identity, the core idea again is that it has something to do with the nature of being necessary.

But we still should be careful on the formulation. For example, $\exists q(Nx \equiv_x \forall p(\Box(p \equiv p) \wedge q))$ might seem natrual; but it would not work. For one thing, it is quite mysterious why would something being necessary involve that any fact of generalized identity is necessary. It also does not yield $\Box(p \equiv p)$, for it is consistent with there being no necessary truths.

Instead, I suggest the following: $\exists q(([\lambda G.Gx \equiv_x Nx]F) \equiv_F (\forall p \Box(p \equiv p) \wedge q))$. This, however, is not to say it is true in virtue of the nature of being necessary that $\Box(p \equiv p)$; but rather that it is true by the nature of the what is it to be *identical with the property of being necessary*. In other words, the objectual essence of the property of being necessary essentially yields that $\Box(p \equiv p)$. In this case, it is not involved in something being necessary that any fact of generalized identity is necessary; and $\Box(p \equiv p)$ is guaranteed for $Ns \equiv_s Ns$ is a theorem. Now essentialists can proceed their argument as it was. We can by substitutability get $\Box(p \equiv (q \wedge r))$ and $\Box(p \equiv q)$, which means that the fact on what is essential to what is necessary.

To sum up, although I admit there might be a gap between essence and necessity as skeptics

challenge, essentialists can still fill in the gap by appealing to the essence of being necessary. I argued that similar challenge could be raised for *any* non-transmissivists, and they should appeal to the essence of being necessary to answer the challenge. The nature of necessary should be enough to guarantee (GNP). In case of definitional essentialists, they can directly take (NP) as being true in virtue of the nature of being necessary. The case is more complex for who understands essence in terms of generalized identity, but still it is possible to take $\Box(p \equiv p)$ to be true in virtue of the identity of being necessary.

5 Circularity

I argued in the previous section that the only viable option for non-transmissivists is to take the relevant instances of (GNP) to be true in virtue of the nature of being necessary. This was done by showing that any other alternatives are incompatible with non-transmissivism. But to say that is only viable option for non-transmissivists is not to say it is teneble. After all, non-transmissivism might also incompatible with the thesis that the relevant instance of (GNP) is true in virtue of the nature of being necessary; and with the argument I presented it completes a trilemma which likely undermines any kind of non-transmissivism.

It seems that there are no *logical* inconsistency with the thesis and non-transmissive explanation on necessity. To say the nature of being necessary is such and such is one thing, and to say the necessity of necessary truths is explained by such and such factor is another. Similar to essentialism, which is a instance of non-transmissivism.

But it could be the case that the thesis seriously undermines the theoretical purpose of non-transmissivism. The reason to refuse to think that (GNP) is true in virtue of the nature of the criteria *C* was not that there is a logical inconsistency. The argument against it was that it results in the disturbing consequence of *C*, which was meant to explain necessity, being ontologically dependent on being necessary.

The thesis that (NP) is true in virtue of the nature of being necessary would not face the problem of fundamentality. By including the notion of essence, or being essentially true into

the essence of being necessary neatly place non-modal notion of essence in a more fundamental position than being necessary.

But perhaps there are other problems. Romero (2019) states six constraints on the notion of explanation which the ‘explanation’ of modality by essence must satisfy:

Why The explanation must answer to a ‘why’ question.

Acyclicity The explanation should not be circular.

Fundamentality The explanans should be metaphysically more fundamental than the explanandum.

Ontic and Factive It must be the case that the truth of the explanans, not our concept or theory of it, explains the truth of the explanandum.

Structural The explanandum need not be an object or a property.

Objectivity The explanation should be objectively valid. (p.124)

The discussion in section 3 would made it clear that my suggestion does not make essentialism infringe **Fundamentality**. Essentialism also easily satisfies **Why**, **Ontic and Factive**. It answers the question “*Why is p necessary?*”, by appealing to the *facts* on essence of things, not our *concept* of essence. Essentialists would certainly hold that if the essentialist explanation on necessity is valid, then it is *objectively* valid. Finally, **Structural** is not a negative constraint that one can infringe.

However, whether my suggestion is compatible with the constraints on **Acyclicity** is perhaps not so obvious. Some might feel uncomfortable to the idea that the nature of being necessary validates the explanation from essence to necessity. But if it is indeed circular, what exact part of the picture contains a circle?

As a reminder, the outline of the picture I am suggesting is as follows. First there is essentialist thesis: for any necessary truth¹⁰ p , $\Box p$ because p is essential to some objects. It would be

¹⁰Alternatively, any fundamental necessary truth.

obvious that then the notion of essence should not be understood in terms of necessity. But then there is a gap. Essentialists want to hold that, on the one hand the necessity of *every* necessary truths can be such explained, and on the other hand we do not need additional explanatory components to account for the necessity. But then essentialists commit themselves into the thesis that every propositional true in virtue of the nature of some objects are necessary - i.e., (NP). But how can essentialists guarantees (NP), if they understand the notion of essence *without* the notion of necessity? And the answer for that is my final thesis: (NP) is true in virtue of the nature of *being necessary*.

Notice that I do not try to explain the necessity of p by appealing to the essence of being necessary. The explanatory work is done by the fact that p is true in virtue of the nature of some objects. The urge to account for the implicit assumption on (NP) arises only when essentialists are asked how *all* necessities are explained by essence. The thesis that (NP) is true in virtue of the nature of being necessary explains, indirectly, why we can generalize the point. In particular cases of the explanation of the source of necessity, the nature of being necessary remains irrelevant.

Hence there is at least no blatant circularity. Actually, it is even unclear whether it would be viciously circular to involve the essence of being necessary in particular explanations. Suppose I am wrong in the previous paragraph, and every explanation on necessity must include the fact that (NP) is true in virtue of the nature of being necessary. p is necessary because p is true in virtue of the nature of some objects *and every such truths are necessary in virtue of the nature of being necessary*. What is wrong with this kind of explanation? My point perhaps can be made by a simple analogy. If someone asks me why 3 is a prime number, I can simply answer “because 3 is a natural number that is not a product of two natural numbers smaller than itself.” However, if someone thinks this is incomplete, then I can add: “because 3 is a natural number that is not a product of two natural numbers smaller than itself, and every such natural numbers are prime in virtue of the nature of being prime”. It would seem obvious that both explanation for 3 being prime are not viciously circular.

More intriguing problems regarding circularity target (NP) specifically. It may seem natural

to suppose (NP) itself is also necessary. But needs to be clarified whether essentialists are really committed into the necessity of (NP). I have formulated the account in a way that avoids such a commitment. Essentialists must hold that every truths that obtains in virtue of the nature of some *objects* are necessary. However, my suggestion was that (NP) is the *predicational* essence of being necessary, not an objectual essence of, for example, the logical object which is the necessity operator. Thus by placing (NP) into the essence of being necessary does not force essentialists to admit that (NP) is necessarily true.

However, there certainly are some circumstances where essentialists are automatically committed into the necessity of (NP). One obvious case is that if one insists the (NP) is true in virtue of the nature of the logical object of necessity. In this case (NP) itself commands (NP) to be necessary. Simillar point can be made if one understands essence in terms of generalized identity, and prefers to take $\Box(p \equiv p)$ to be the partial essence of the identity of being necessary. This is the option considered in the end of the fourth section. If an essentialists prefers that option, by the necessity of generalized identity one would committed into the necessity of $\exists q(([\lambda G.Gx \equiv_x Nx]F) \equiv_F (\forall p\Box(p \equiv p) \wedge q))$.

Also when an essentialist wishes to include the essence of being necessary in all explanation on source of necessity. Here the explanans of the necessity of p would be the conjunction of the essentiality of p and the essentiality of (NP). Since essentialists try to give a non-transmissive explanation, the conjunction should also be necessary, although the explanation does not appeal to the necessity. And since for a conjunction to be necessarily true each conjuncts should be necessarily true; it follows that it should be necessary that (NP) is true in virtue of the nature of being necessary. And it would be obvious that if it is necessarily true that (NP) is true in virtue of the nature of being necessary, then (NP) is necessary.¹¹

Now let us temporarily grant that (NP) is indeed necessary, and therefore by essentialism (NP) is true in virtue of the nature of some objects, for example the necessity operator, or the

¹¹ Actually, this is quite a disturbing consequence because it is *not* a theorem in the logic of essence that $\Box_F A \rightarrow \Box_F \Box_F A$, where F is a non-rigid predicate such as “being necessary”. But also not plausible that $\Box_F A$ to be true in virtue of the nature of other objects, which is required for it to be necessary. I think the suggestion that the essence of being necessary should be involved in source of all necessity goes better rather with the suggestion that (NP) is true in virtue of some *objects* related to necessity, such as the operator $\Box$. Then (NP) would be true in virtue of the nature of the *rigid* predicate $\lambda x.x = \Box$; and $\Box_F A \rightarrow \Box_F \Box_F A$ is a theorem where F is a rigid predicate.

object which is the property of being necessary. Let such object be n . Now, it does not pose a serious problem if (NP) is the essence independently of each n and being necessary. But perhaps it is also plausible to suppose that (NP) being the essence of n grounds (NP) being the essence of being necessary. Therefore, (NP) holds ultimately because of the nature of n , and my suggestion collapses to the thesis that the gap between essence and necessity is filled by the nature of n .

The puzzle is then as follows. Essentialists will account the source of $\Box(\text{NP})$ as $\Box_n(\text{NP})$. Now why does $\Box_n(\text{NP})$ grants $\Box(\text{NP})$? It might seem there is no other choice to appeal to the fact that (NP) is true in virtue of the nature of n . That is: $\Box_n(\text{NP})$ grants $\Box(\text{NP})$ because $\Box_n(\text{NP})$. This is outright circularity, or is just to say there is no gap at all rather than to fill in the gap. And I admitted that there is a gap.

One answer to the alleged puzzle is, of course, to deny (NP) is necessarily true. Notice that by doing so one do *not* abandon the necessity of necessity: $\Box p \rightarrow \Box\Box p$. This is again because $\Box_F A \rightarrow \Box_F \Box_F A$ is a theorem where F is rigid. But even we grant the necessity of (NP), I think there is a possible reply. Remind the argument from $\Box_\alpha \Phi \alpha$ to $\Box \Phi \alpha$ in section 4. The essentiality of (NP) is indeed one crucial premises of the argument, but not the sole premise. The proof utilized T -axiom of the logic of essence and the inference rule Modus Ponens and Universal instantiation. And it is quite customary to suppose the instances of logical axioms and inference rule is true in virtue of the nature of the relevant logical notions (Fine 1994, 1995a). Therefore, one possible answer to the puzzle is to say, in the most rigorous sense, the answer to the problem of why $\Box_n(\text{NP})$ grants $\Box(\text{NP})$ should appeal to not only the essence of n but also the essentialist operator, conditional and universal quantifier.

One might feel uncomfortable to include the essence of essentialist operator in the discourse of necessity. For one thing, did not I rejected to regard (NP) being true in virtue of the essence of essence? Yes, but to regard the “modal axioms”, that is, axioms governing the essentialist operator involve the notion of necessity or possibility. So here the problem of fundamentality does not occur. Even if we identify the notion of essence and the Finean essentialist operator, it does not follow that essence should be understood in terms of necessity, and therefore es-

sentialism is circular. It nevertheless does suggest that essence is ontologically dependent on logical connectives, set-theoretical notions, or some notions represented by the metalanguage of logic. This, however, is what essentialists *can* admit. Fine himself, for instance, clarifies his position as holding essence as “methodologically primitive”, which is “perfectly compatible with its not actually being a primitive.” (Fine 2020, p.465)

6 Conclusion

I argued that essentialists must accept that (NP) is true in virtue of the nature of being necessary. This follows from the more general character of non-transmissivism. Not only essentialist theories on the source of necessity, but all non-transmissivist theories face the problem of the gap. When one asserts that p is necessary because $C(p)$, one has to elucidate the notion of C without modal notions, and at the same time grant (GNP): $\forall p(C(p) \rightarrow \Box p)$. I argued in section 3 that all non-transmissivists should regard (GNP) to be true in virtue of the nature of being necessary. And since all essentialist theories are non-transmissivist - this is argued in section 1 - essentialists are forced to conclude (NP) is true in virtue of the nature of being necessary. I also tried to show that this account is not circular as it may first appear.

It is worth noting that this seems to yield the conclusion that essentialists cannot hold the notion of being necessary to be metaphysically fundamental. It should be ontologically dependent to the notion of being essentially true of something. Therefore, if Hale meant metaphysical fundamentality by “fundamental and irreducible” character of necessity, his position seems no longer tenable. As long as he does not argue for the implausible view that necessity is independent of the predication of being necessary, either he must abandon essentialist theory on the source of necessity or abandon the fundamentality of necessity.

What I did *not* argue in this paper is that essentialism is superior to other competing theories. My aim in this paper is, as I have written, that the particular challenge to essentialists can be met in a coherent way without dismissing it. Although the skeptical challenge is correct, it is not strong enough to defeat essentialism. Therefore the conclusion of the paper is somewhat

merely conditional: *if* essentialism is correct, then (NP) is true in virtue of the nature of being necessary. I mentioned that any argument in favor of essentialism is also an argument for the essentiality of (NP). But of course the contrapositive holds. Any criticism against (NP) would thereby be a criticism towards essentialism. And dealing with those criticisms would exceed the scope of this paper.

References

- Blackburn, S. (1993). Morals and modals. In S. Blackburn (Ed.), *Essays in quasi-realism* (pp. 52–73). Oxford University Press.
- Cameron, R. P. (2010). On the source of necessity. In B. Hale & A. Hoffmann (Eds.), *Modality: metaphysics, logic, and epistemology* (pp. 137–152). Oxford University Press.
- Correia, F., & Skiles, A. (2017). Grounding, essence, and identity. *Philosophy and Phenomenological Research*, 98(3), 642–670. doi: 10.1111/phpr.12468
- Fine, K. (1982). The problem of non-existents: I. internalism. *Topoi*, 1(1-2), 97–140. doi: 10.1007/bf00157548
- Fine, K. (1994). Essence and modality. *Philosophical Perspectives*, 8(Logic and Language), 1–16. doi: 10.2307/2214160
- Fine, K. (1995a). The logic of essence. *Journal of Philosophical Logic*, 24(3), 241–273. doi: 10.1007/bf01344203
- Fine, K. (1995b). Ontological dependence. *Proceedings of the Aristotelian Society*, 95(1), 269–290. doi: 10.1093/aristotelian/95.1.269
- Fine, K. (1995c). Senses of essence. In W. Sinnott-Armstrong, D. Raffman, & N. Asher (Eds.), *Modality, morality, and belief: essays in honor of ruth barcan marcus* (pp. 53–73). Cambridge University Press.
- Fine, K. (2015). Unified foundations for essence and ground. *Journal of the American Philosophical Association*, 1(2), 296–311. doi: 10.1017/apa.2014.26
- Fine, K. (2020). Comments on penelope mackie’s ”can metaphysical modality be based on essence?”. In M. Dumitru (Ed.), *Metaphysics, meaning, and modality: Themes from kit*

- fine* (pp. 461–465). Oxford University Press.
- Glazier, M. (2017). Essentialist explanation. *Philosophical Studies*, 174(11), 2871–2889. doi: 10.1007/s11098-016-0815-z
- Hale, B. (2002). The source of necessity. *Noûs*, 36(s16), 299–319. doi: 10.1111/1468-0068.36.s16.11
- Hale, B. (2013). *Necessary beings: An essay on ontology, modality, and the relations between them*. Oxford, England: Oxford University Press.
- Leech, J. (2018). Essence and mere necessity. *Royal Institute of Philosophy Supplement*, 82, 309–332. doi: 10.1017/s1358246118000139
- Leech, J. (2021). From essence to necessity via identity. *Mind*, 130(519), 887–908. doi: 10.1093/mind/fzaa012
- Lowe, E. J. (2008). Two notions of being: Entity and essence. *Royal Institute of Philosophy Supplement*, 62, 23–48. doi: 10.1017/s1358246108000568
- Lowe, E. J. (2012). What is the source of our knowledge of modal truths? *Mind*, 121(484), 919–950. doi: 10.1093/mind/fzs122
- Mackie, P. (2020). Can metaphysical modality be based on essence? In M. Dumitru (Ed.), *Metaphysics, meaning, and modality: Themes from Kit Fine* (pp. 247–264). Oxford University Press.
- Priest, G. (2005). *Towards non-being: The logic and metaphysics of intentionality*. New York: Oxford University Press.
- Romero, C. (2019). Modality is not explainable by essence. *Philosophical Quarterly*, 69(274), 121–141. doi: 10.1093/pq/pqy043
- Zalta, E. N. (1983). *Abstract objects: An introduction to axiomatic metaphysics*. Dordrecht, Netherland: D. Reidel.
- Zalta, E. N. (2006). Essence and modality. *Mind*, 115(459), 659–693. doi: 10.1093/mind/fzl659